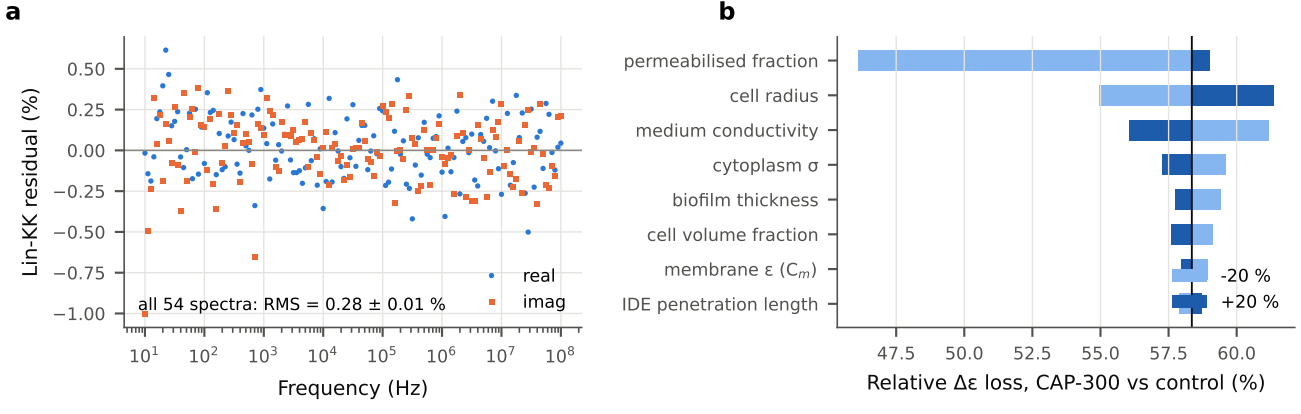
a
Lin-KK residual (%)
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
all 54 spectra: RMS = 0.28 ± 0.01 %
real
imag
10^1 10^2 10^3 10^4 10^5 10^6 10^7 10^8
Frequency (Hz)
b
permeabilised fraction
cell radius
medium conductivity
cytoplasm σ
biofilm thickness
cell volume fraction
membrane ε (C_m)
IDE penetration length
-20 %
+20 %
47.5 50.0 52.5 55.0 57.5 60.0
Relative Δε loss, CAP-300 vs control (%)